Making LoRa APRS a little smarter

Ideas borrowed from Meshtastic, and what Richard Guzman's iGate firmware already has

Part 1. What Meshtastic does that LoRa APRS doesn't

This part is from memory of Meshtastic's design, not rechecked against current docs; some details have changed between versions.

Meshtastic	LoRa APRS today
Neighbor discovery: every node broadcasts a periodic NodeInfo packet (name, hardware, role). Each node keeps a table of who it has heard, with SNR and RSSI.	A station beacons its position and comment, but nobody keeps a neighbor table, and there is no capability announcement.
SNR-based rebroadcast: a node waits a random delay that is shorter the weaker the packet was (a far node repeats first). If it hears someone else repeat the packet during that wait, it cancels.	A digipeater repeats `WIDE1-1` almost at once. Two digipeaters that both hear a packet can collide. The iGate firmware does have channel-activity-detection backoff.
Implicit acknowledgement: when a node hears a neighbor rebroadcast its packet, it knows the packet got out. It retries if it never hears that.	Beacons have no confirmation. Only directed messages get an ACK.
Hop limit in a header, and duplicate suppression: every packet carries an ID, and nodes drop repeats.	`WIDEn-N` already limits hops. Many digipeaters drop duplicates, but not consistently.
Learned next-hop routing for directed messages, so they stop flooding.	Messages go as floods or via fixed digipeater paths.
Channel-utilization limits: nodes back off when the channel is busy.	Beacon timing ignores how busy the channel is.
Roles (client, router, mute) that change how a node behaves.	Fixed roles by convention only.

Things that need no on-air change (easy wins)

- **SNR-weighted digipeat delay with cancel-on-hear.** Local to the digipeater, and the biggest collision reducer.
- **Implicit ack for beacons.** A tracker listens for its own callsign coming back with a digipeated mark in the path (`CALL*`), then retries or raises power if it doesn't hear that.
- **Airtime-aware beaconing.** Stretch the interval when the channel is busy.
- **A heard-stations table** with RSSI and SNR, shown on the iGate's web page.

Things that need a small format addition

- **Capability announcement.** One status or beacon marker that says "I decode compressed frames, I'm a digipeater, my hop policy is X." This is what would make a listen-and-switch tracker work reliably.
- **Learned next hops for messages,** using the last-heard path, with the existing message-number ACK.

Things to leave alone

- **Encryption.** Meshtastic's channel encryption doesn't fit amateur radio rules, which require content to be readable and stations to identify with a callsign.

- **A new addressing scheme.** Callsigns are how APRS-IS, aprs.fi and every map work. Extend APRS rather than becoming Meshtastic.

Risk

Flooding scales badly. Meshtastic itself has had to add hop-limit changes and role restrictions to keep large meshes from jamming. Copying the neighbor discovery and the SNR delay is low risk. Copying general flood routing is not.

Suggested start: SNR-weighted digipeating and implicit acks. Both live inside Richard's iGate and tracker firmware, need no on-air format change, and are easy to test against real traffic.

Part 2. What Richard's iGate firmware already has

Checked by reading the code in `LoRa_APRS_iGate` (`lora_utils.cpp`, `digi_utils.cpp`, `station_utils.cpp`, `map_utils.cpp`).

Already there

- **Channel-activity-detection backoff before transmitting.** It waits for the channel to be clear (`waitForDIFS`, 2 slots) and then waits a random 1-4 slots (`doBEB`) before sending. This is Wi-Fi-style listen-before-talk. It reduces collisions between nearby transmitters, but it isn't the SNR-weighted scheme.
- **Duplicate suppression.** `digi_utils.cpp` drops a packet if the same sender and payload were seen in the last 25 seconds (`isIn25SegHashBuffer`).
- **A heard-stations table with signal quality.** `map_utils.cpp` keeps up to 50 stations with RSSI, SNR, hit count and a `heardDirect` flag (arrived without a digipeater star in the path). Entries expire after an hour and show on the web map.
- **A "recently heard" list.** `station_utils.cpp` tracks last-heard stations with a timeout. The iGate uses it to decide whether to transmit messages to a station (`wasHeard`).
- **Echo avoidance.** It ignores its own packets and won't repeat an answer to a message addressed to itself.

Not there

- **SNR-weighted digipeat delay with cancel-on-hear.** Digipeated packets go into an output queue and are sent in order, with only the CAD backoff and the 25 s duplicate filter in the way. Signal strength isn't used to decide who repeats first, and a pending repeat isn't cancelled when another digipeater sends the same packet.
- **Implicit acks for beacons.** There is no check that a station's own packet came back digipeated.
- **Capability announcement.** Nothing tells neighbors what a station supports.
- **Airtime-aware beaconing.** I saw no logic that stretches intervals when the channel is busy, though I only read the files above, so I could have missed some.

He has the groundwork: a heard table with signal quality, and a basic collision guard. The first two items in the easy-wins list would be new, and the data they need is already collected.

I did not read the LoRa_APRS_Tracker repo, so I can't say what it does.